

WHITE PAPER

Smart Industrial Distribution Board with Fault Recording Capability

An intelligent distribution board that monitors electrical parameters in real time and records fault data to reduce downtime and improve maintenance efficiency.

THE PROBLEM

Industrial facilities rely on continuous and reliable electrical power to maintain production. However, traditional distribution boards only indicate that a fault has occurred without providing information about the electrical conditions before the fault. This increases troubleshooting time, causes production delays, and raises maintenance costs.

As industrial systems become more complex, there is a growing need for smart monitoring solutions that can quickly identify faults and provide maintenance teams with accurate diagnostic information.

OUR SOLUTION

The Smart Industrial Distribution Board with Fault Recording Capability continuously monitors voltage, current, power consumption, and panel temperature using intelligent sensors and an Arduino-based controller.

When a fault occurs, the system automatically records the latest operating conditions, identifies the affected circuit, and generates a fault report. This built-in "black box" functionality enables faster fault diagnosis and significantly reduces maintenance downtime.

PROOF IT WORKS

- Voltage monitoring accuracy greater than 95%
- Current monitoring accuracy greater than 95%
- Fault identification within 2 seconds
- Successful storage and display of fault data after simulated fault events

WHAT WE'RE LOOKING FOR

We are seeking industrial partners, factories, maintenance companies, and investors interested in piloting, validating, and expanding this technology for real-world industrial applications.

HOW IT COMPARES

Feature	Our Solution	Traditional Board	Advanced Systems
Real-Time Monitoring	✓ Yes	✗ No	✓ Yes
Fault Recording	✓ Yes	✗ No	✓ Yes
Fault Identification	✓ Yes	Limited	✓ Yes
Installation Cost	Low	Low	High
Maintenance Support	Excellent	Basic	Excellent
IoT Ready	✓ Yes	✗ No	✓ Yes

HOW IT WORKS

System at a Glance



Figure 1: Smart Industrial Distribution Board Architecture

The 4-Step Process

1	Data Acquisition: Voltage, current, and temperature sensors continuously collect electrical measurements from the distribution board.
2	Real-Time Processing: The Arduino controller processes the incoming sensor data and calculates power consumption in real time.
3	Fault Detection & Recording: When a fault signal is detected, the system identifies the affected circuit and stores the latest operating measurements.
4	Reporting & Analysis: A fault report is generated and displayed, allowing maintenance personnel to quickly analyze and diagnose the issue.

Key Technologies

Arduino Uno

Voltage Sensors

Current Sensors

Temperature Sensor

LCD Display

Fault Recording System

Electrical Protection Devices

Embedded C/C++

Arduino IDE

Future IoT Integration

RESULTS & VALIDATION

≥95% VOLTAGE ACCURACY	≥95% CURRENT ACCURACY	≤2s FAULT DETECTION TIME
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The prototype was tested using simulated fault conditions to ensure safe operation without creating actual electrical hazards. During testing, the system continuously monitored voltage, current, power, and temperature while displaying real-time measurements on the LCD display.

When a fault simulation event was triggered, the system successfully identified the affected circuit, recorded the latest operating conditions, and generated a fault report. The recorded information proved valuable for fault diagnosis and maintenance analysis. The results demonstrate the feasibility of integrating fault-recording capabilities into industrial distribution boards to improve reliability and maintenance efficiency.

GET INVOLVED

We are looking for industry partners, investors, and collaborators to take this solution from prototype to deployment.

FOR INDUSTRY Pilot With Us Deploy the prototype in a real industrial environment and evaluate its effectiveness in reducing maintenance downtime.	FOR FUNDERS Fund the Next Phase Support the development of a full-scale industrial version with advanced monitoring and IoT connectivity.	FOR RESEARCHERS Collaborate With Us Partner with our team to enhance fault analytics, remote monitoring, and predictive maintenance capabilities.
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CONTACT THE TEAM

Role	Details
Supervisor	Dr. Mohamed Mandour Department: Electrical Power Engineering Email: Mohomed.mandor@sut.edu.eg
Project Page / Repo	https:// (QR Code for QMS)

Team Members & Bios

Name / ID	Role	Contribution & Background
Hamza Ahmed Youssef ID: 2401032585	Team Leader	Coordinated project planning, system integration, and documentation.
Mohamed Awwad Zaki ID: 240102987	Logistics Coordinator	Managed project resources, procurement, and scheduling activities.
Mustafa Ahmed Youssef ID: 240103257	Software Engineer	Developed embedded software and monitoring functions.
Abdelrahman Hamde Ahmed ID: 240102795	Problem Solving Engineer	Responsible for system analysis and troubleshooting strategies.
Mahmoud Essam Rabia ID: 240103443	Hardware Engineer	Designed hardware circuits and system wiring.
Menna Mohamed Shapan ID: 230100057	Software Engineer	Assisted in programming, testing, and software validation.
Antonions Essam Saad ID: 240100988	Hardware Engineer	Assisted in hardware assembly and testing procedures.
Omar Mohamed Abdelmonem ID: 240103399	Energy Engineering Student	Contributed to electrical system design, power monitoring analysis, testing, and validation of the Smart Industrial Distribution Board.

This project was developed at El Sewedy University of Technology (SUT) — Egypt's polytechnic university dedicated to applied engineering and industry collaboration. SUT students work on projects connected to real industrial partners across the energy, manufacturing, and technology sectors.

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